

ECON 0100 | Homework B2 | Supply

| Solutions

Answers and work are shown in the red blocks.

Butterbeer

The supply curve for butter beer can be represented by the following equation:

$$P = 20 + \frac{1}{2}Q$$

Prices are in galleons and quantity is in bottles.

Q1 | Quantity Supplied

Use a graph to plot this supply curve, including the quantity supplied at both 40 galleons and 50 galleons.

a) Quantity supplied at 40 galleons: _____

40 bottles.

Plot. Vertical intercept, $Q = 0$: $P = 20$. Below 20 galleons quantity supplied is zero, so the curve starts there and rises by 1 galleon for every 2 bottles.

$$40 = 20 + \frac{1}{2}Q \rightarrow 20 = \frac{1}{2}Q \rightarrow Q = 40$$

b) Quantity supplied at 50 galleons: _____

60 bottles.

$$50 = 20 + \frac{1}{2}Q \rightarrow 30 = \frac{1}{2}Q \rightarrow Q = 60$$

c) How much did quantity supplied change as the price dropped from 50 to 40? _____

Down by 20 bottles.

■ $\Delta Q_s = 40 - 60 = -20$: quantity supplied fell by 20 bottles as the price fell, the law of supply.

d) What is marginal cost at a quantity of 60? _____

50 galleons.

Quantity to price. The height of supply at 60 bottles is the marginal cost of the sixtieth bottle:
 $P = 20 + \frac{1}{2} \cdot 60 = 50$ galleons. It matches b): 50 galleons is the price at which 60 bottles are supplied.

Q2 | Producer Surplus

Then find and label the producer surplus at these prices.

a) PS at 40 galleons: _____

400 galleons.

Producer surplus is the triangle above supply and below the price, out to the quantity sold from Q1. Its height is the price minus the 20-galleon intercept and its base is the quantity.

$$P = 40 : \quad PS = h \cdot b \cdot \frac{1}{2} = (40 - 20) \cdot 40 \cdot \frac{1}{2} = 400$$

b) PS at 50 galleons: _____

900 galleons.

$$P = 50 : \quad PS = h \cdot b \cdot \frac{1}{2} = (50 - 20) \cdot 60 \cdot \frac{1}{2} = 900$$

c) How much did producer surplus change as the price dropped from 50 to 40? _____

Down by 500 galleons.

$\Delta PS = PS - PS = 400 - 900 = -500$: producer surplus fell by 500 galleons, the strip between the two price lines.

Revenue is the rectangle below the price: $40 \cdot 40 = 1600$ at 40 galleons and $50 \cdot 60 = 3000$ at 50 the rest of each rectangle covers marginal costs.

